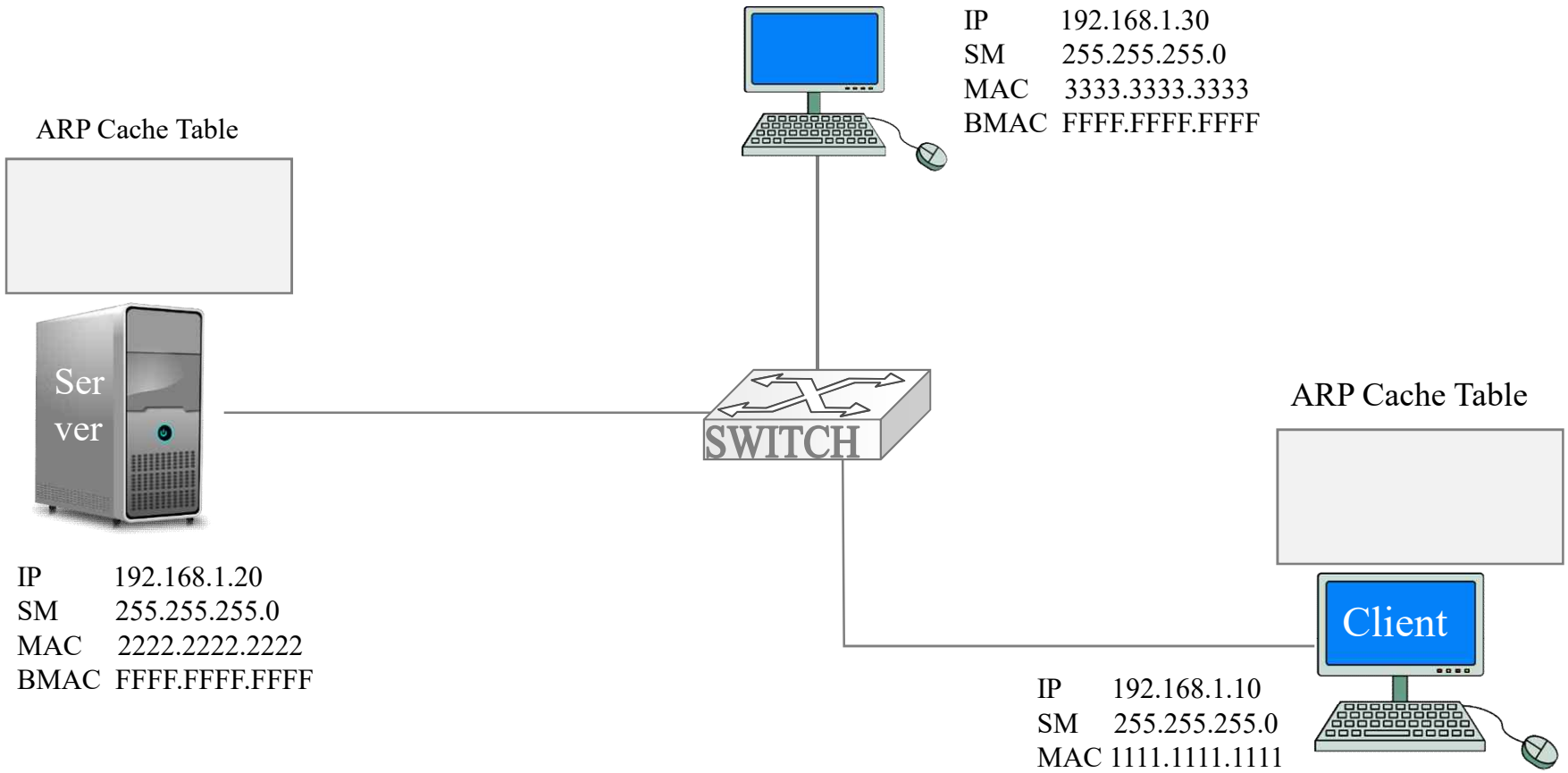
ARP cache table update



ARP Request

수MAC	송MAC	Protocol	ARP Header	
FFFF.FFFF.FFFF	1111.1111.1111	0x0806	1111.1111.1111 /192.168.1.10 **0000.0000.0000/192.168.1.20**	송신지 수신지

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
????	1111. 1111.1111	192.168.1.10	**192.168.1.20**	50030	80	Get /www.test.com

	ARP Header	Protocol	송MAC	수MAC
송신지 수신지	2222.2222.2222 /192.168.1.20 1111.1111.1111 /192.168.1.10	0x0806	2222.2222.2222	1111.1111.1111

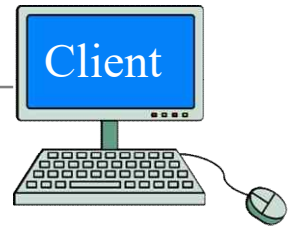
ARP Cache Table

192.168.1.10	1111.1111.1111

IP 192.168.1.20
SM 255.255.255.0
MAC 2222.2222.2222
BMAC FFFF.FFFF.FFFF

ARP Cache Table

192.168.1.20	2222.2222.2222



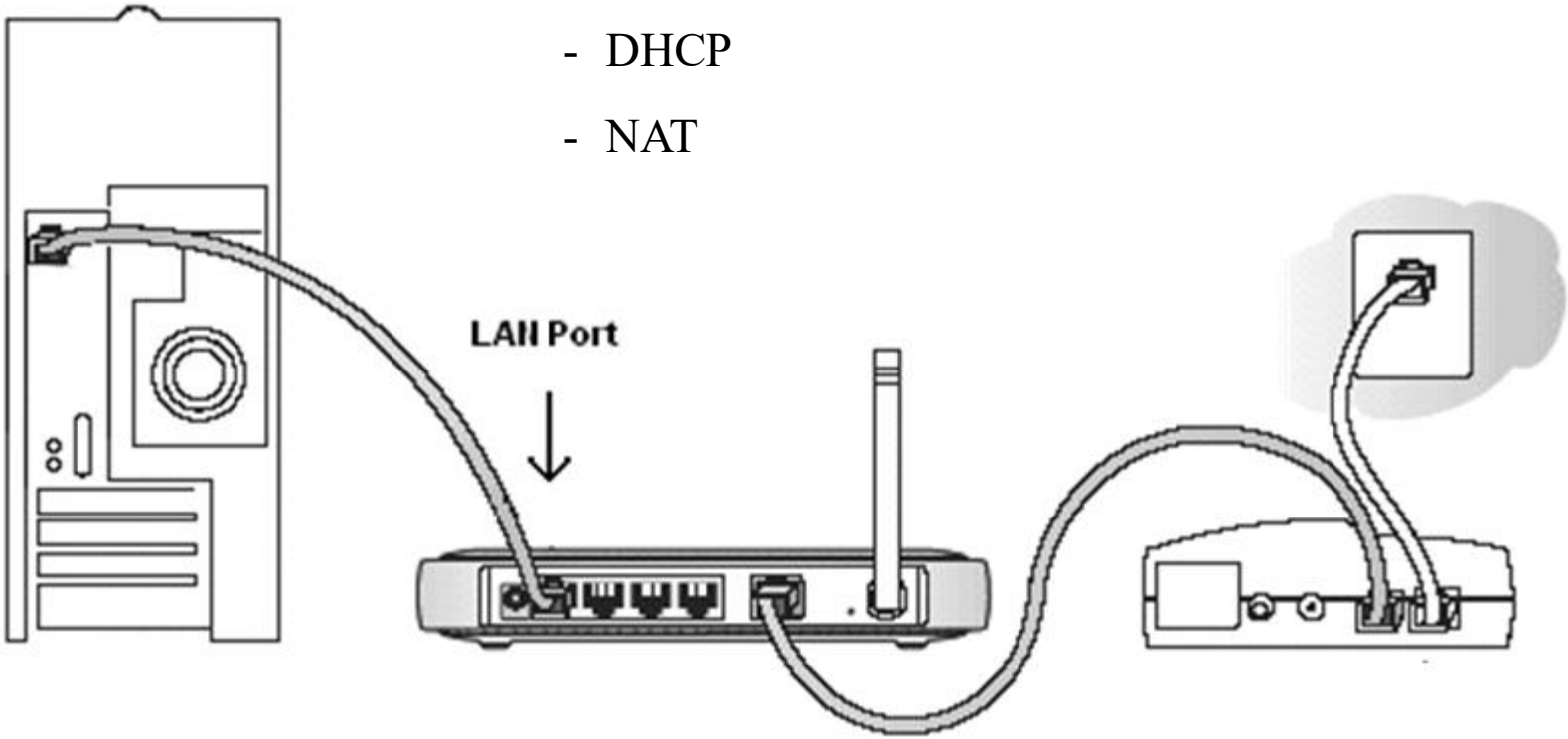
IP 192.168.1.10
SM 255.255.255.0
MAC1111.1111.1111

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
222.2222.2222	1111. 1111.1111	192.168.1.10	192.168.1.20	50030	80	Get /www.test.com

NAT
(Network Address Translation)

SoHo(Small office/home office)에서의 NAT

- 공유기 기능
  - Forwarding
  - Routing
  - DHCP
  - NAT



❶ 컴퓨터 ❷ 유/무선 공유기 ❸ 모뎀

NAT 기능

- 네트워크 주소 변환 (Network Address Translation)
- 송신지 또는 수신지 IP address를 다른 주소로 변환
- Source NAT(SNAT)와 Destination NAT(DNAT)

송신지 IP	수신지 IP
192.168.1.10	200.10.10.10

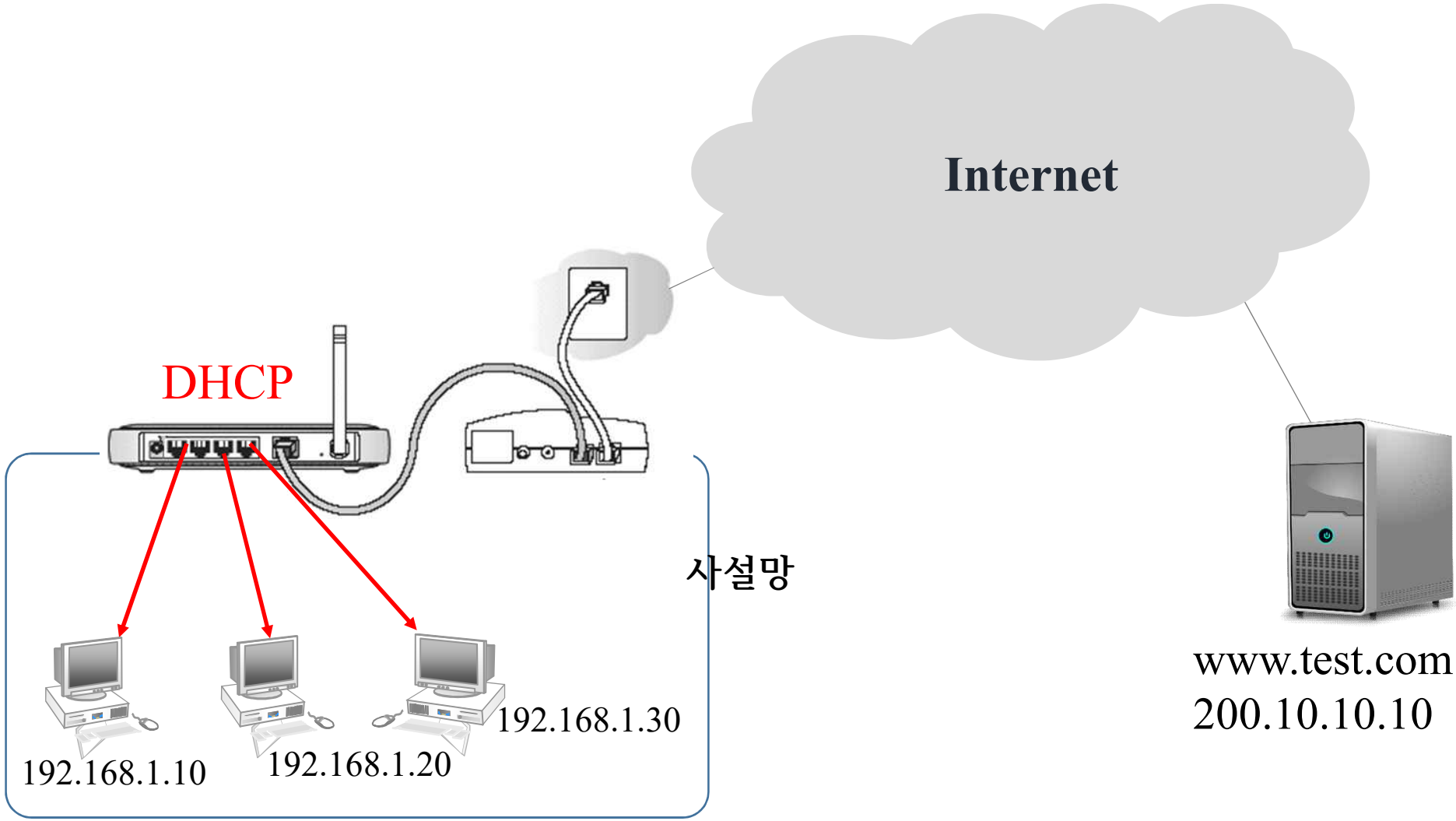
❶ **SNAT**

172.16.1.10	200.10.10.10

송신지 IP	수신지 IP
192.168.1.10	**200.10.10.10**

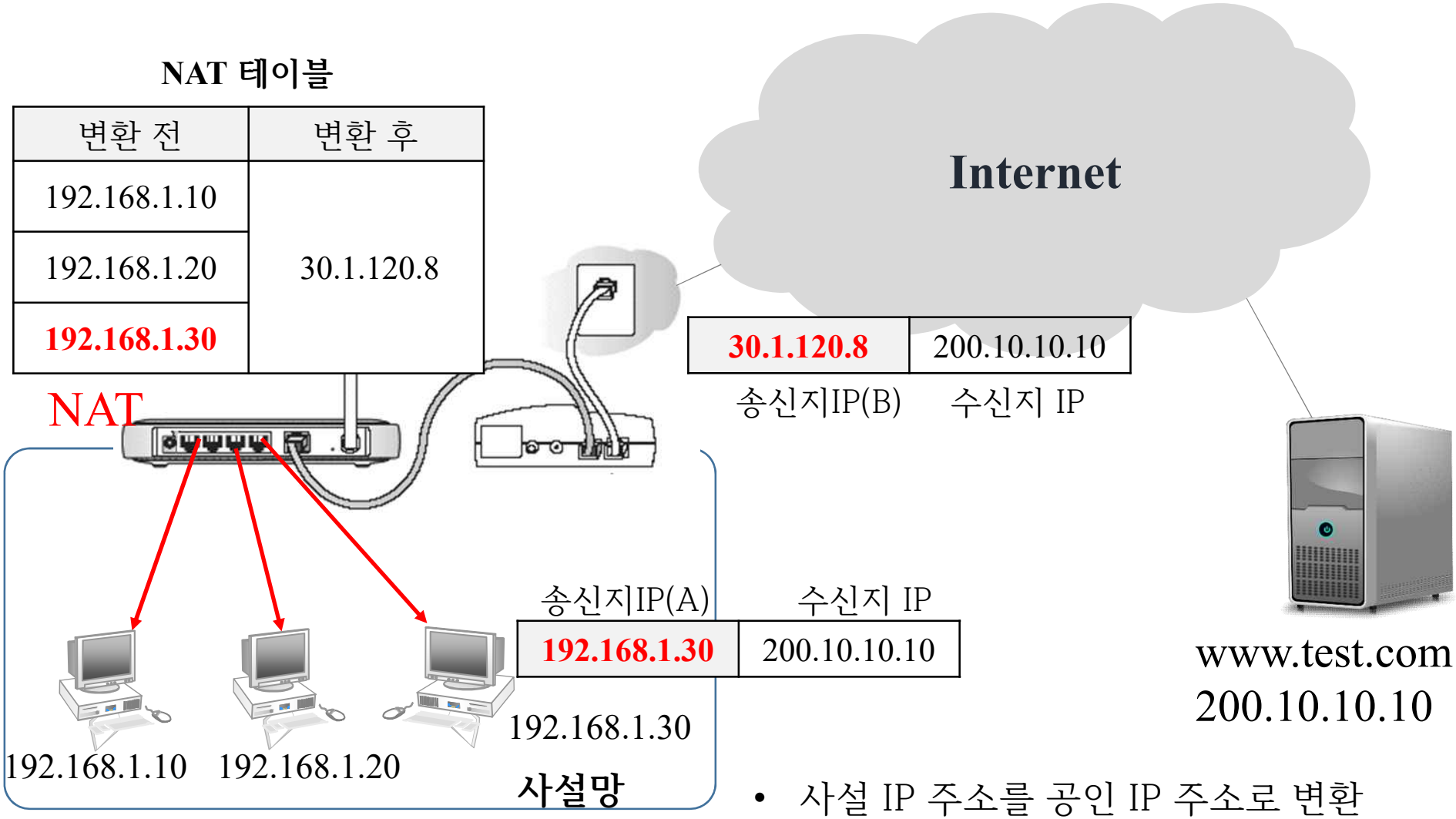
❷ **DNAT**

192.168.1.10	**10.10.10.10**


Internet
DHCP
사설망
192.168.1.10
192.168.1.20
192.168.1.30
www.test.com
200.10.10.10

NAT 테이블

변환 전	변환 후
192.168.1.10	30.1.120.8
192.168.1.20	
192.168.1.30	



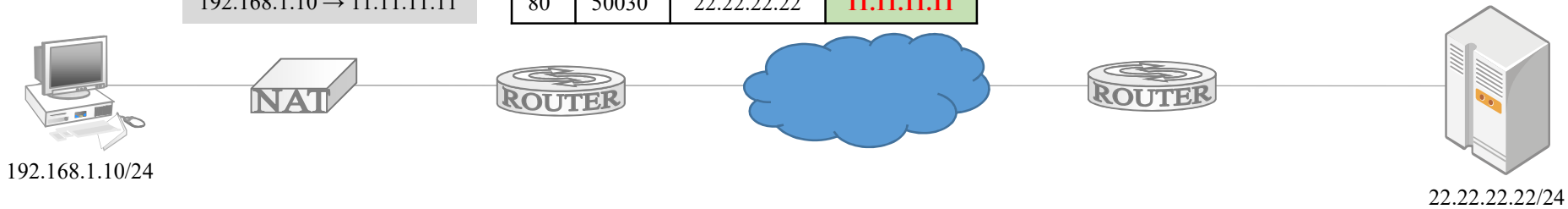
- 사설 IP 주소를 공인 IP 주소로 변환

Source NAT

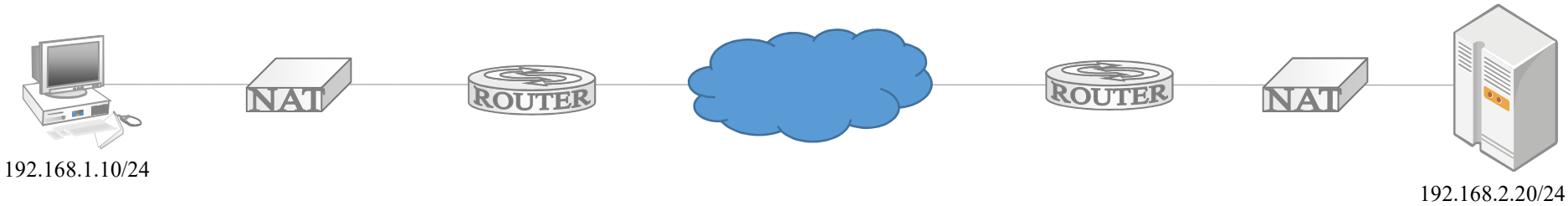
NAT Table

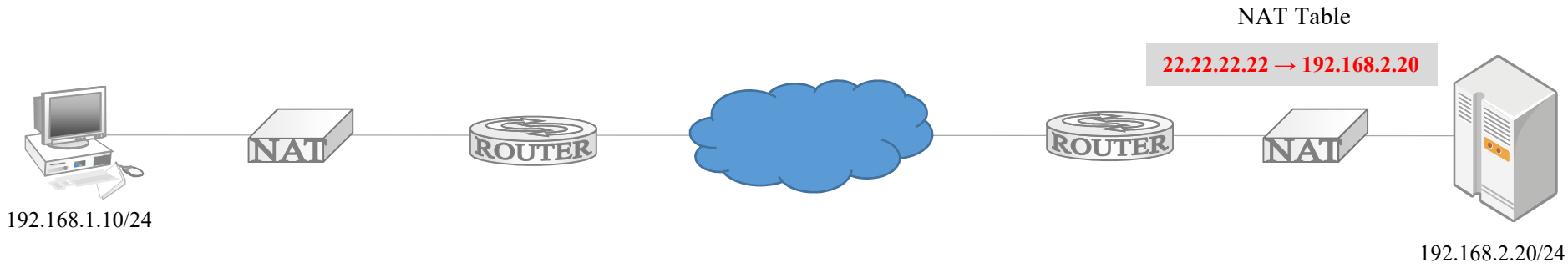
192.168.1.10 → 11.11.11.11

수Port	송Port	수IP	송IP
80	50030	22.22.22.22	11.11.11.11

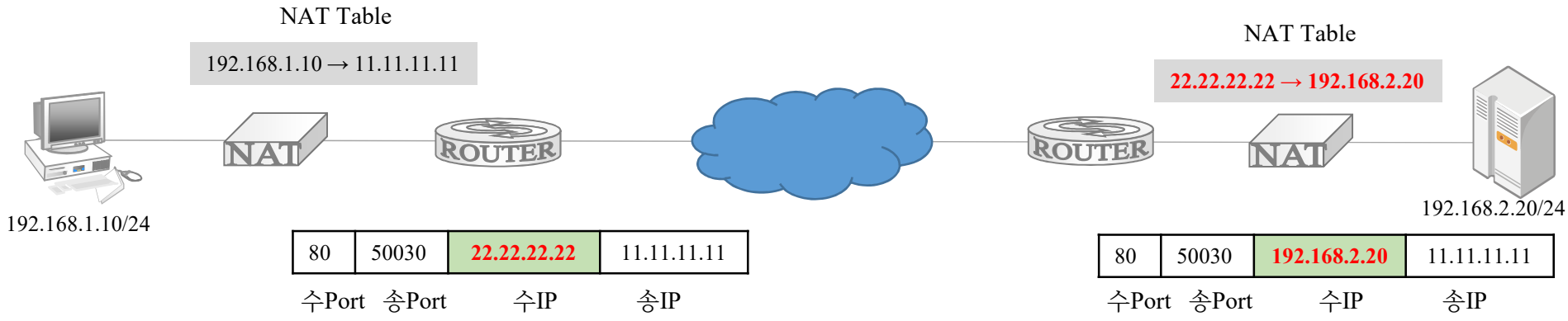


수Port	송Port	수IP	송IP
80	50030	22.22.22.22	192.168.1.10

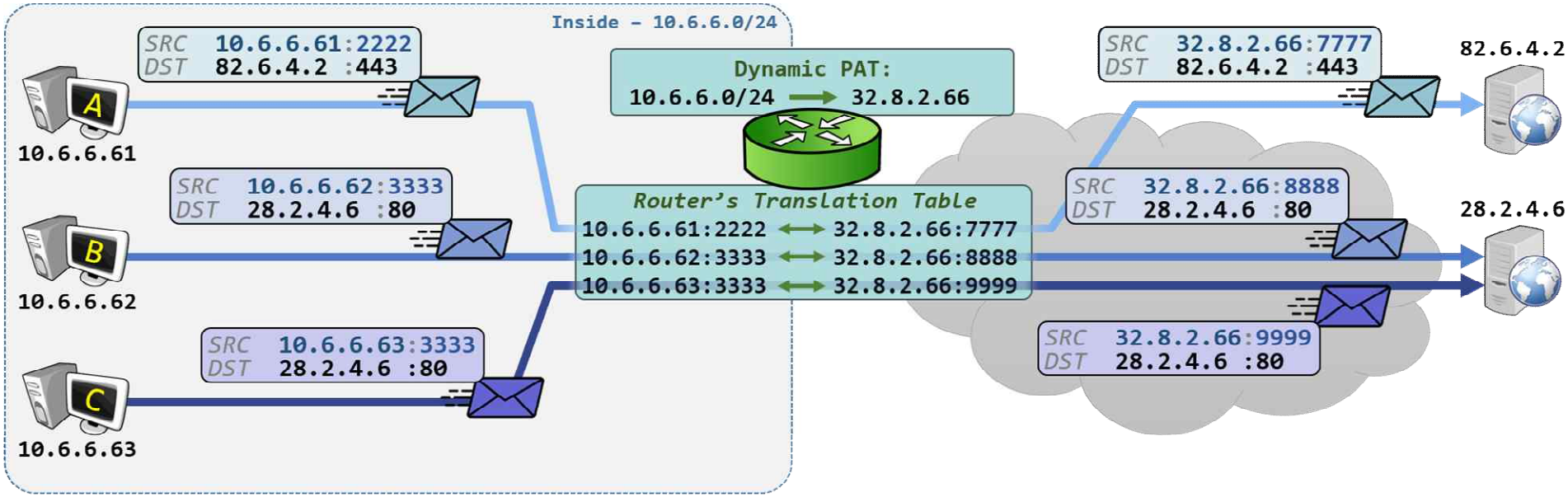


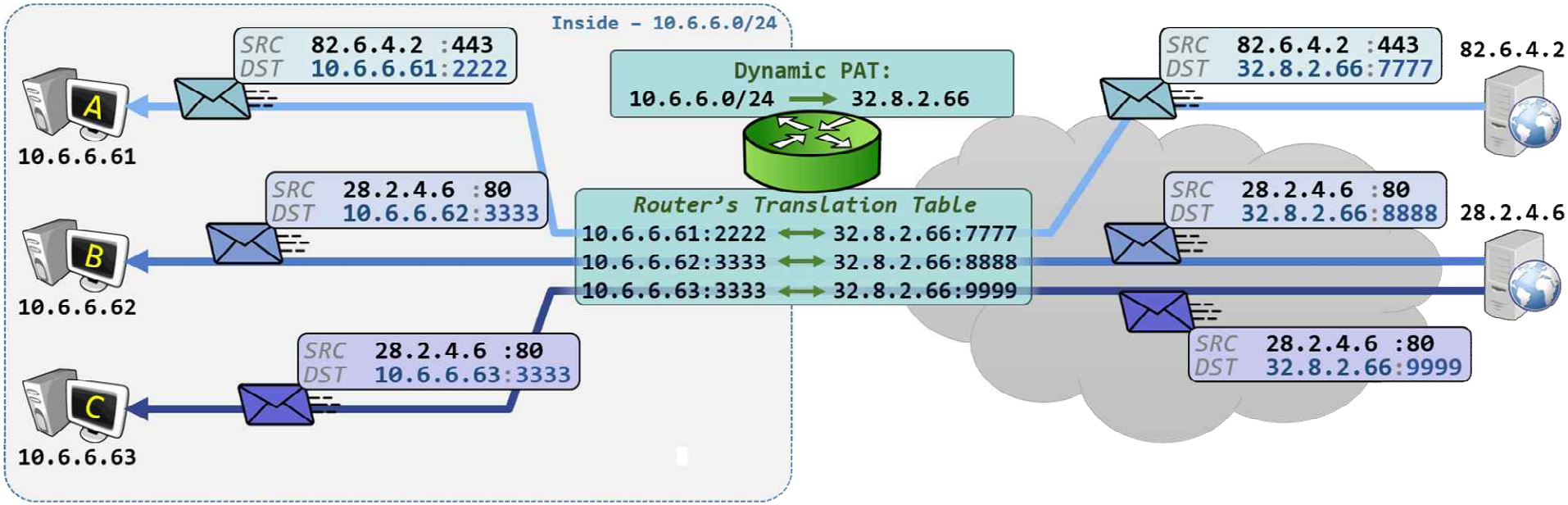


Destination NAT

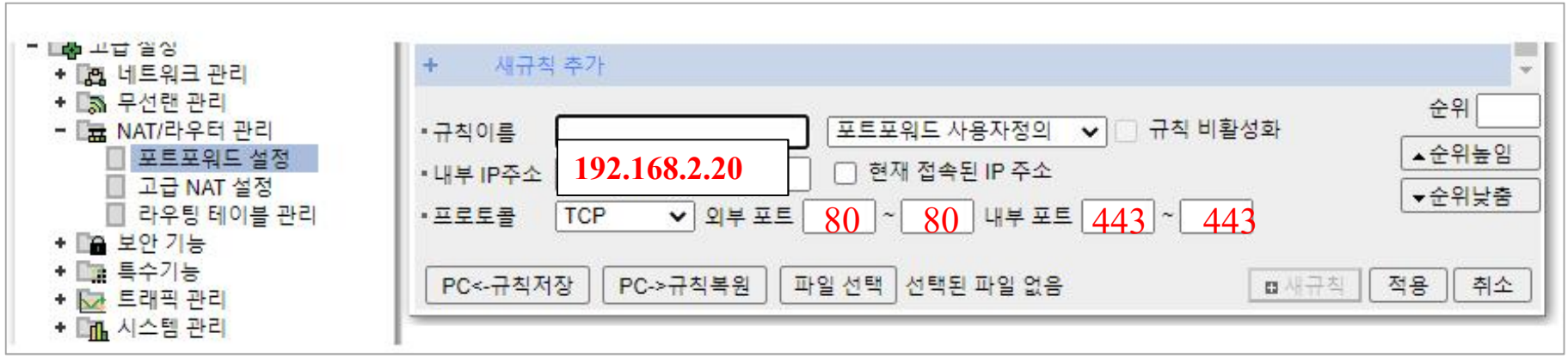


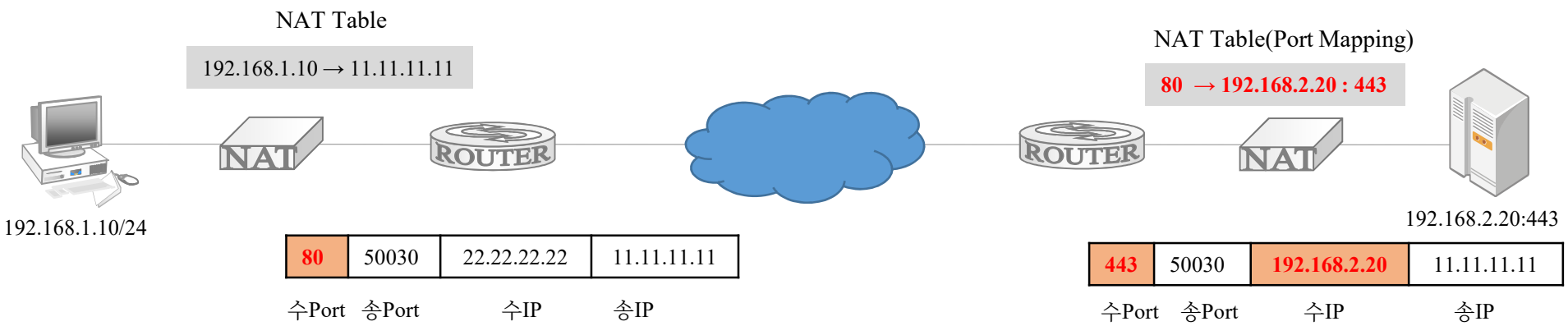
PAT


Inside – 10.6.6.0/24
SRC 10.6.6.61:2222
DST 82.6.4.2 :443
A
10.6.6.61
SRC 10.6.6.62:3333
DST 28.2.4.6 :80
B
10.6.6.62
SRC 10.6.6.63:3333
DST 28.2.4.6 :80
C
10.6.6.63
Dynamic PAT:
10.6.6.0/24 → 32.8.2.66
Router's Translation Table
10.6.6.61:2222 ↔ 32.8.2.66:7777
10.6.6.62:3333 ↔ 32.8.2.66:8888
10.6.6.63:3333 ↔ 32.8.2.66:9999
SRC 32.8.2.66:7777
DST 82.6.4.2 :443
82.6.4.2
SRC 32.8.2.66:8888
DST 28.2.4.6 :80
28.2.4.6
SRC 32.8.2.66:9999
DST 28.2.4.6 :80

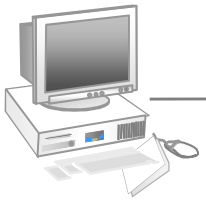

Inside – 10.6.6.0/24
SRC 82.6.4.2 :443
DST 10.6.6.61:2222
Dynamic PAT:
10.6.6.0/24 → 32.8.2.66
SRC 82.6.4.2 :443
DST 32.8.2.66:7777
82.6.4.2
A
10.6.6.61
SRC 28.2.4.6 :80
DST 10.6.6.62:3333
Router's Translation Table
10.6.6.61:2222 ↔ 32.8.2.66:7777
10.6.6.62:3333 ↔ 32.8.2.66:8888
10.6.6.63:3333 ↔ 32.8.2.66:9999
SRC 28.2.4.6 :80
DST 32.8.2.66:8888
28.2.4.6
B
10.6.6.62
SRC 28.2.4.6 :80
DST 10.6.6.63:3333
SRC 28.2.4.6 :80
DST 32.8.2.66:9999
C
10.6.6.63

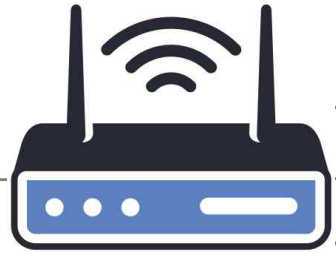
NAT Port Mapping (Port Forwarding)





http://200:1.1.10





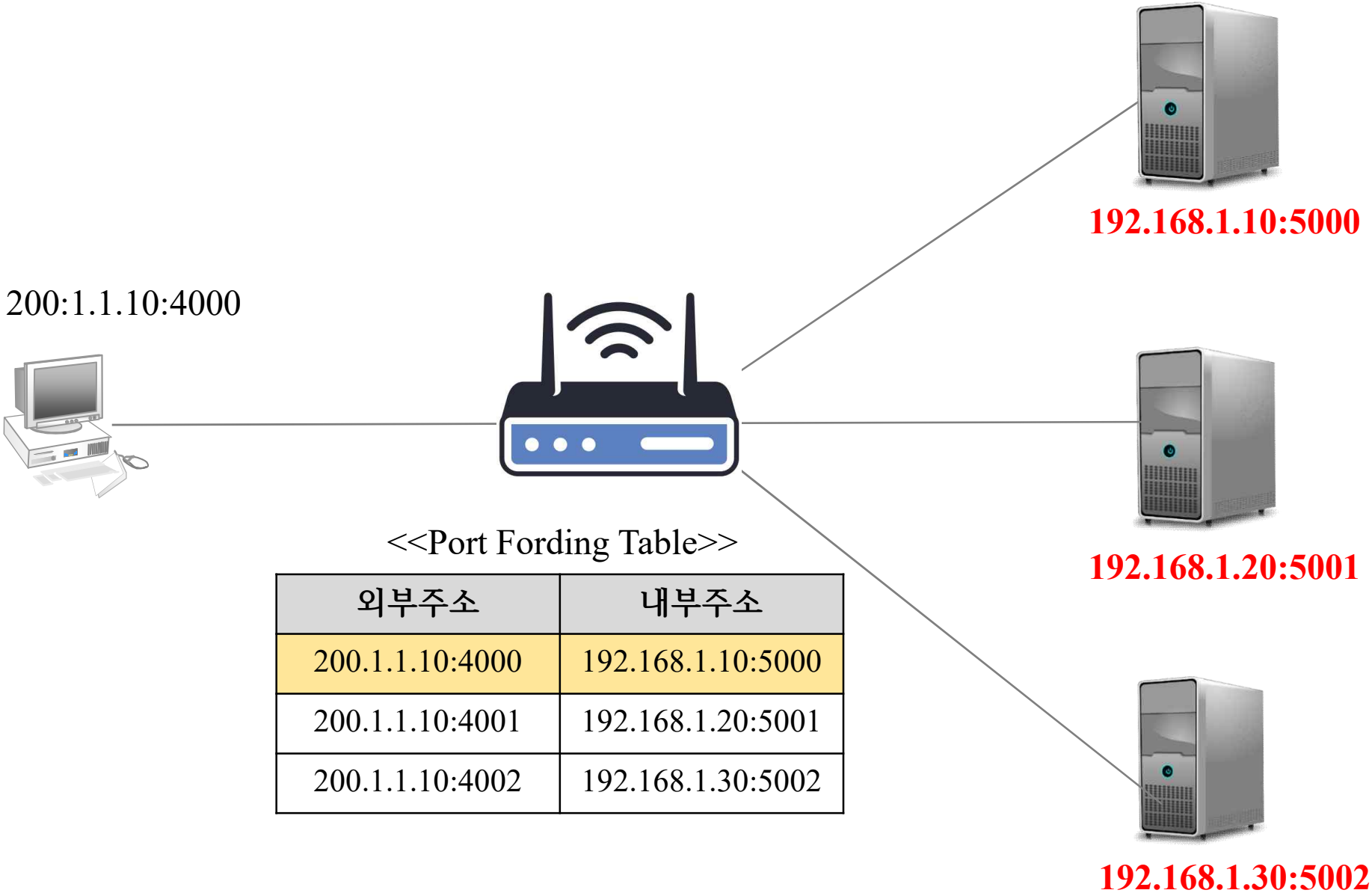
HTTP
192.168.1.10:80

FTP
192.168.1.20:21

RDP
192.168.1.30:3389

<<Port Fording Table>>

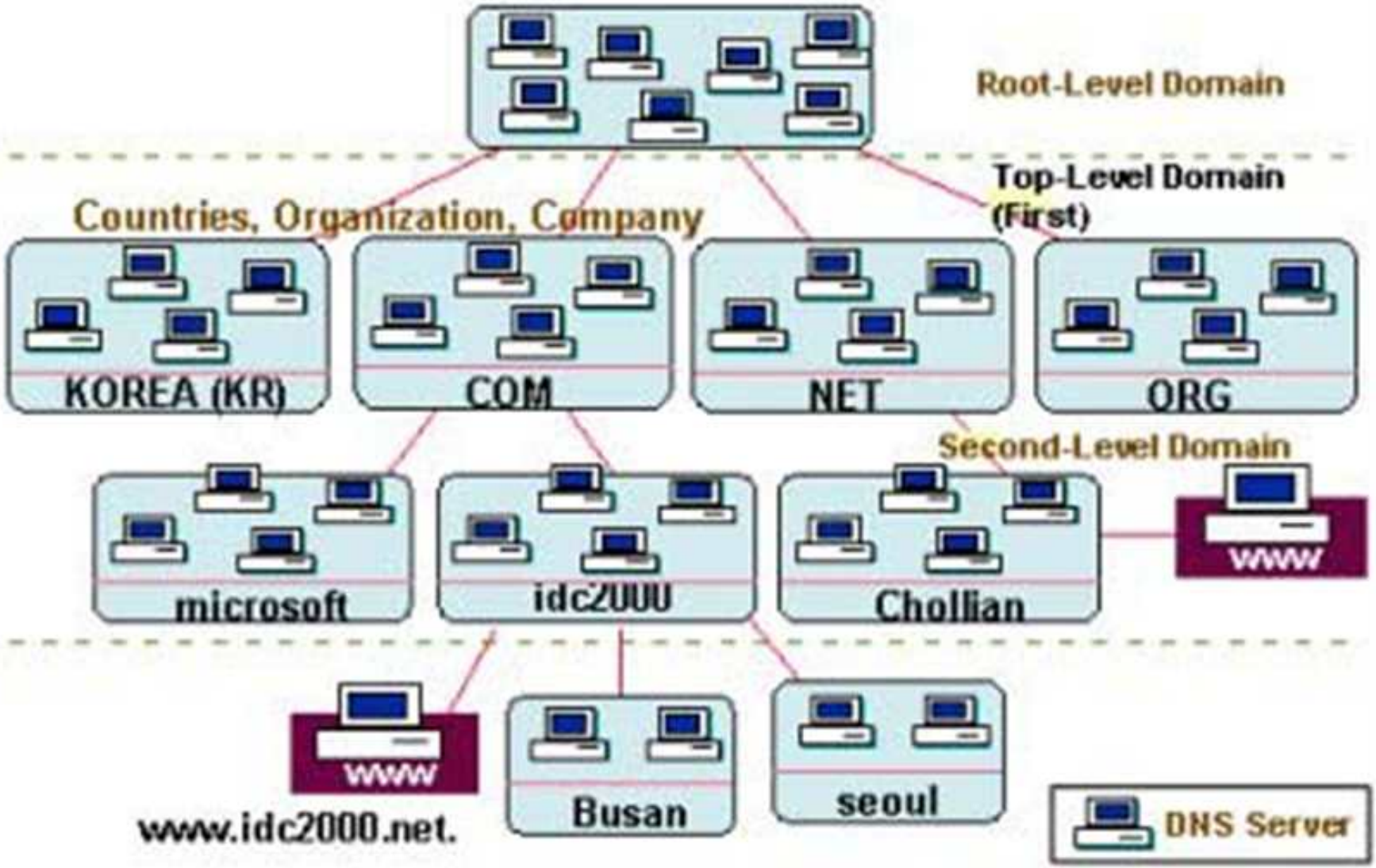
외부주소	내부주소
200.1.1.10:80	192.168.1.10:80
200.1.1.10:21	192.168.1.20:21
200.1.1.10:3389	192.168.1.30:3389



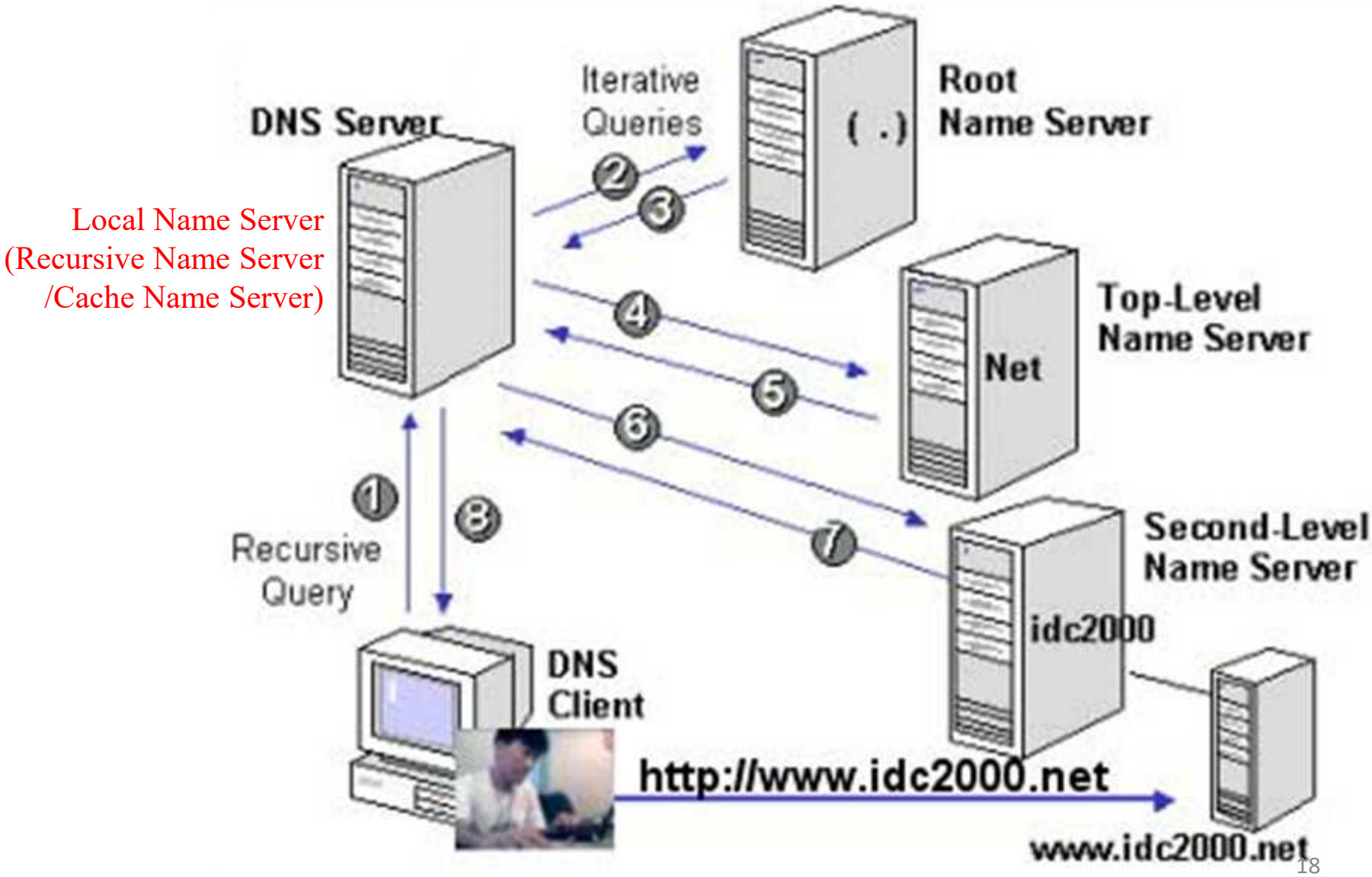
<<Port Fording Table>>

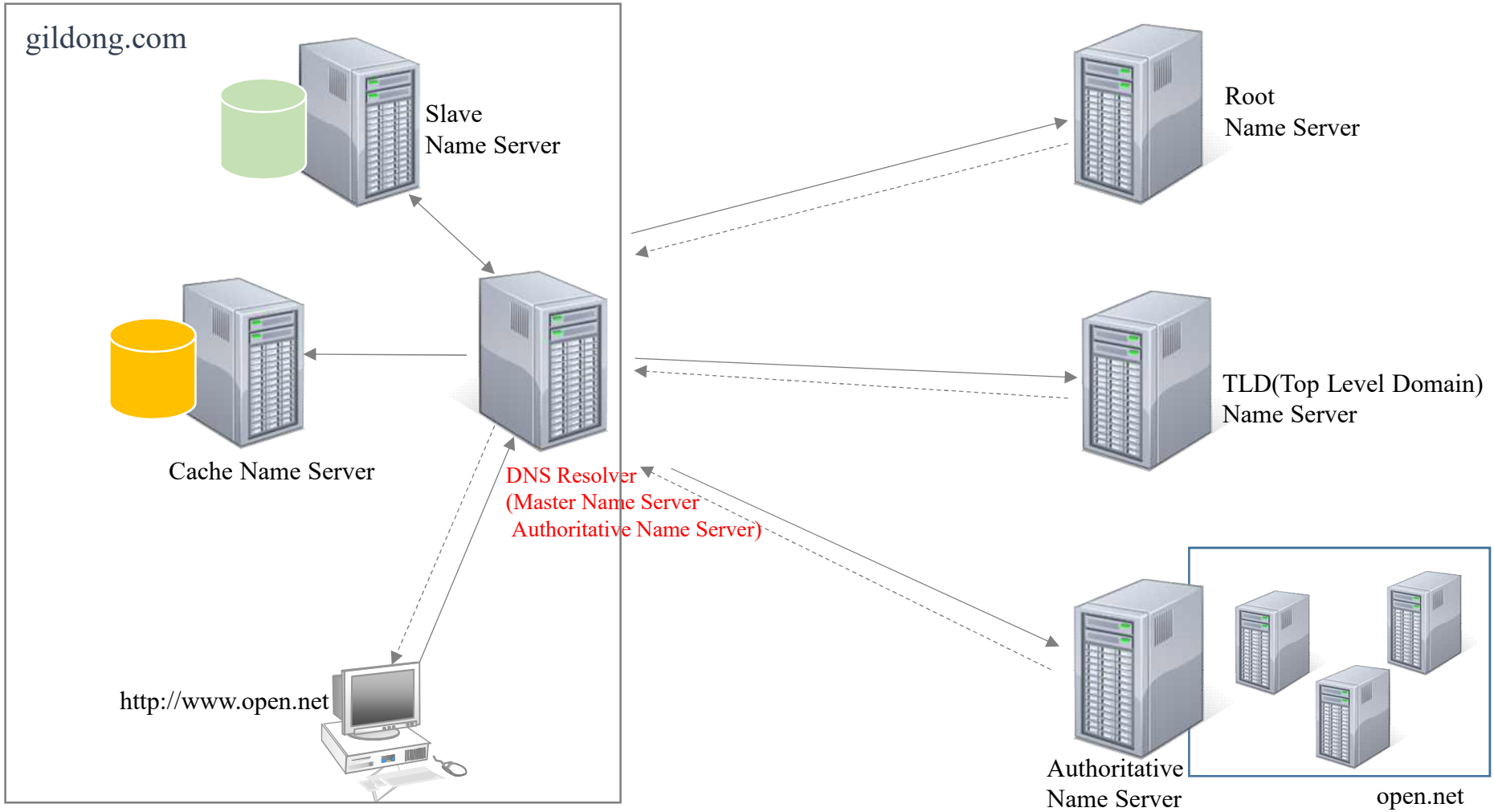
외부주소	내부주소
200.1.1.10:4000	192.168.1.10:5000
200.1.1.10:4001	192.168.1.20:5001
200.1.1.10:4002	192.168.1.30:5002

DNS
(Domain Name Service)

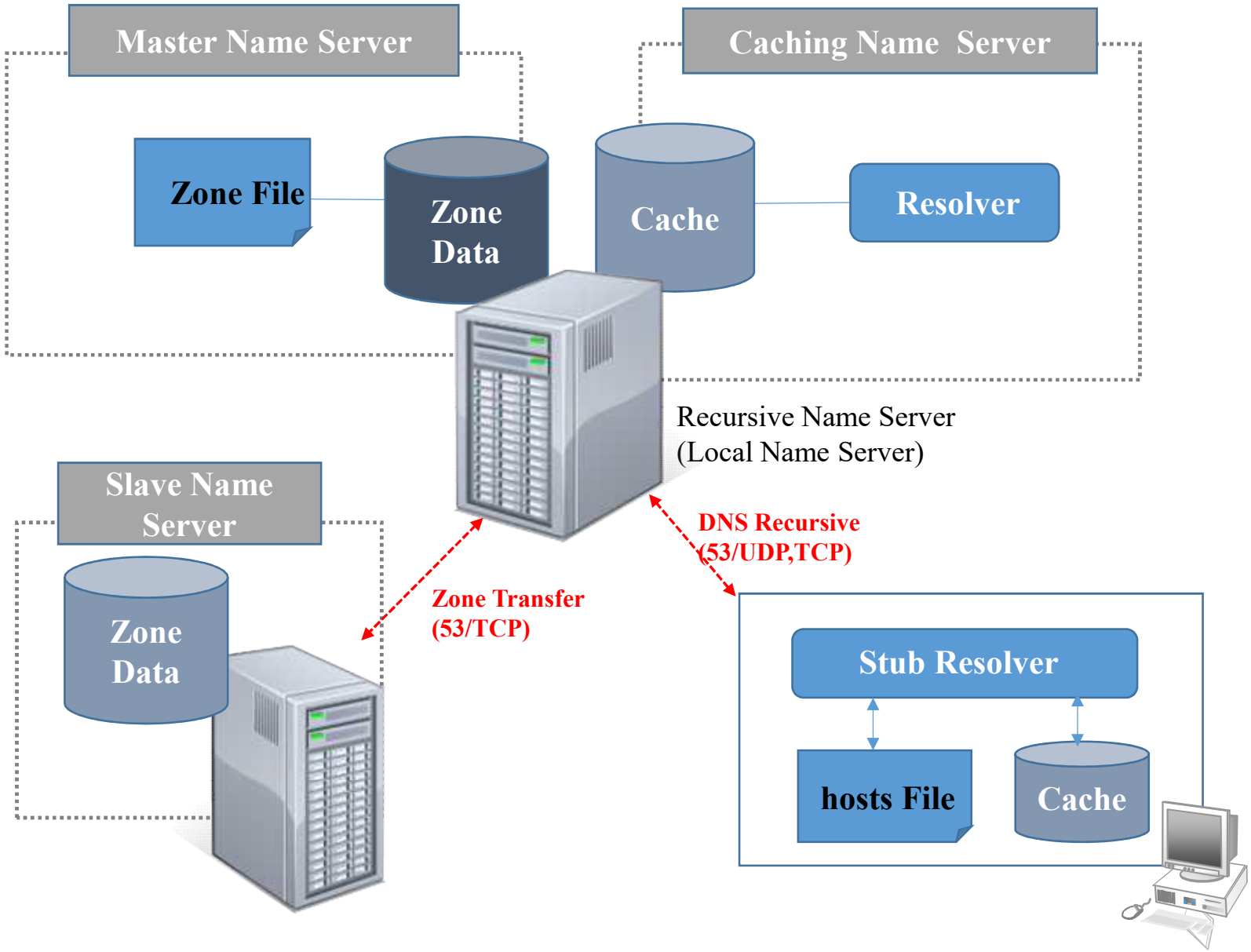

Root-Level Domain
Top-Level Domain
(First)
Countries, Organization, Company
KOREA (KR)
COM
NET
ORG
Second-Level Domain
microsoft
idc2000
Chollian
www
www
www.idc2000.net.
Busan
seoul
DNS Server

DNS Name Resolution




gildong.com
Slave
Name Server
Cache Name Server
DNS Resolver
(Master Name Server
Authoritative Name Server)
http://www.open.net
Root
Name Server
TLD(Top Level Domain)
Name Server
Authoritative
Name Server
open.net

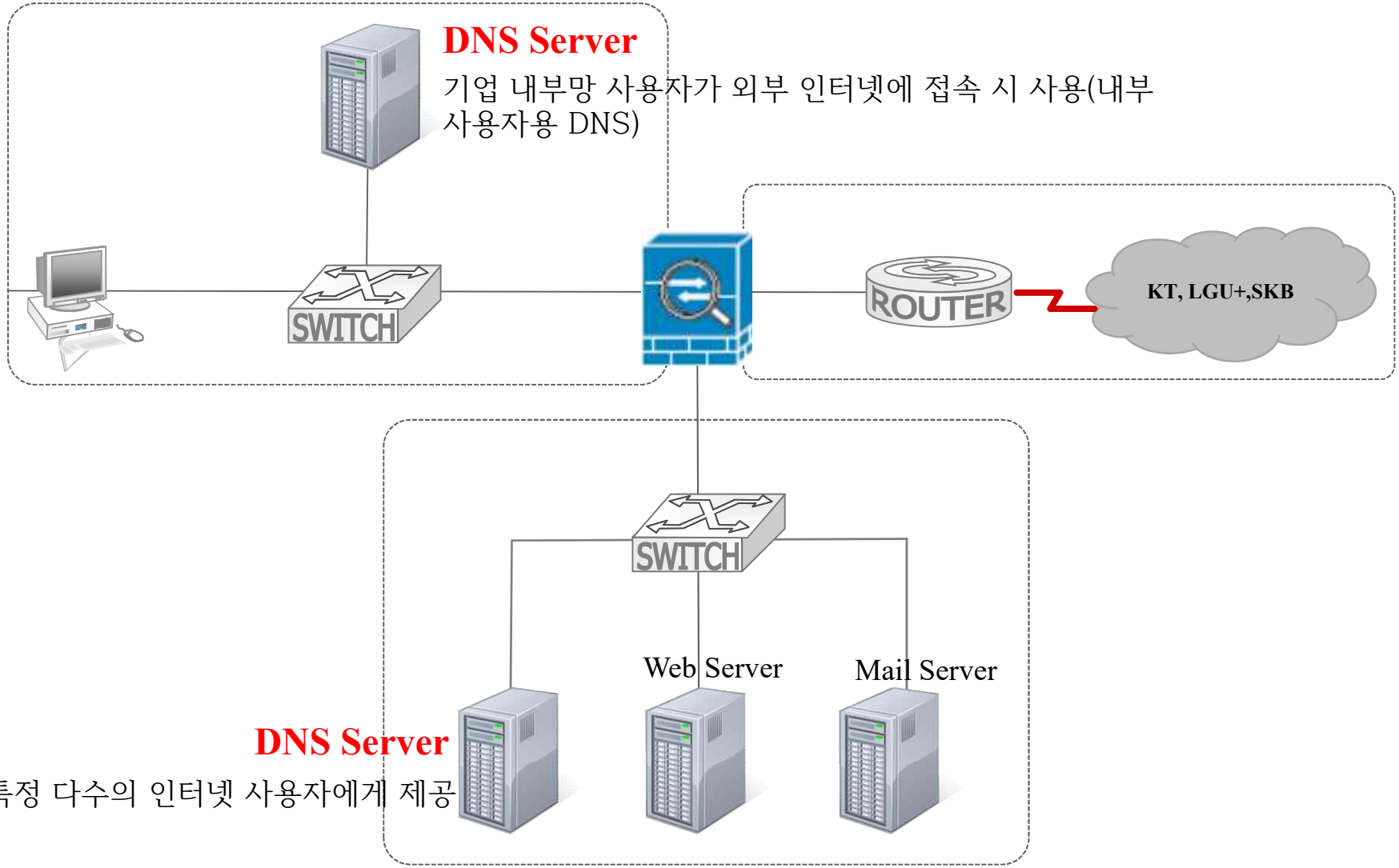
DNS Server


Master Name Server
Zone File
Zone Data
Caching Name Server
Cache
Resolver
Recursive Name Server
(Local Name Server)
Slave Name Server
Zone Data
Zone Transfer
(53/TCP)
DNS Recursive
(53/UDP,TCP)
Stub Resolver
hosts File
Cache

DNS 53 UDP/TCP

- UDP 혹은 TCP를 사용
- 포트번호 : 53
- 응답 메시지의 크기가 512 byte보다 작으면 UDP 사용

  - 최대 UDP 패키지의 크기가 512 byte로 제한되기 때문

  - 응답 메시지의 크기가 512 byte 이상이면 TCP 연결이 사용
- 대부분의 패킷이 UDP를 사용

DNS Server 설치 영역